

George G. Vega Yon, Ph.D.



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Education

Ph.D. in Biostatistics (with a concentration in Statistical Computing)	2020
University of Southern California, USA.	
Dissertation title:	
<i>"Essays on Bioinformatics and Social Network Analysis: Statistical and Computational Methods for Complex Systems."</i>	
M.Sc. in Social Sciences (with a concentration in Economics)	2016
California Institute of Technology, USA.	
Master in Economics and Public Policy	2011
Universidad Adolfo Ibáñez, Chile.	
BS. in Business Administration (with a minor in Political Science)	2011
Universidad Adolfo Ibáñez, Chile.	

Awards

Top paper award, International Communication Association	2022
Travel Grant, Society of Young Network Scientist	2019
Fellowship, California Institute of Technology	2014
Honorable Mention (Posters Session) Chilean Economics Society	2012
Scholarship, Universidad Adolfo Ibáñez	2006



Major Areas of Research Interest

Social Networks and Complex Systems
Statistical Computing
Scientific Software Development
Mechanistic Machine Learning
Statistical Methods Development

Academic and Professional Experience

Research Assistant Professor Division of Epidemiology, University of Utah	Nov 2021 – Present
Data Scientist (contractor) Center for Forecasting and Outbreaks Analytics, CDC	May 2024 – Present
Lead Data Scientist (Associate) Booz Allen Hamilton	Nov 2023 – May 2024
Adjunct Assistant Professor Division of Population Health Sciences, University of Utah	Jan 2023 – Present
Research Programmer Department of Preventive Medicine, University of Southern California	Feb 2018 – Nov 2021
Programmer Analyst Department of Preventive Medicine, University of Southern California.	Oct 2015 – Feb 2018
Graduate student researcher Division of Social Sciences, California Institute of Technology.	Aug 2011 – Oct 2015
Analyst Research Division, Chilean Pension Supervisor (Pension System Watchdog).	Aug 2011 – Aug 2014
Founding partner Nodos Chile Social Network Analysis Ltda.	Jan 2012 – Jan 2014



Peer Reviewed Publications

1. Milando, Chad W. and **Vega Yon, George G.** and Johnson, Kaitlyn and Urbinati, Alessandra and St-Onge, Guillaume and Klein, Brennan and Cori, Anne and White, Laura F.. "A Vision for Estimation of the Instantaneous Reproductive Number." *Epidemics*, pp. 100885. (2026). URL: <https://www.sciencedirect.com/science/article/pii/S1755436526000010> DOI: [10.1016/j.epidem.2026.100885](https://doi.org/10.1016/j.epidem.2026.100885)
2. Sarah Piombo and **George G. Vega Yon** and Thomas W. Valente. "The impact of social norms on diffusion dynamics: A simulation of e-cigarette use behavior." *Health Education & Behavior*. (2025).
3. Sargent, Matthew and Matthews, Luke J. and **Vega Yon, George G.** and Storholm, Erik D. and Ober, Allison J. and Green, Harold D.. "Assessing the Dynamics of PrEP Adoption in a National-Scale Physician Network." *Social Networks* **78**, pp. 226–237. (2024). DOI: [10.1016/j.socnet.2024.02.001](https://doi.org/10.1016/j.socnet.2024.02.001)
4. **Vega Yon, George G.** "Power and Multicollinearity in Small Networks: A Discussion of Tale of Two Datasets: Representativeness and Generalisability of Inference for Samples of Networks" by Krivitsky, Coletti, and Hens." *Journal of the American Statistical Association* **118**(544), pp. 2228–2231. (2023). DOI: [10.1080/01621459.2023.2252041](https://doi.org/10.1080/01621459.2023.2252041)
5. Meyer, Derek and **Vega Yon, George G.** "epiworldR: Fast Agent-Based Epi Models." *Journal of Open Source Software* **8**(90), pp. 5781. (2023). DOI: [10.21105/joss.05781](https://doi.org/10.21105/joss.05781)
6. Panahi, Samin and Kennedy, Eamonn and Roghani, Ali and **Vega Yon, George G.** and VanCott, Anne and Gugger, James J. and Raquel Lopez, Maria and Jo Pugh, Mary. "Veteran Perspectives of Epilepsy Care: Impact of Veteran Satisfaction, Knowledge, and Proactivity." *Epilepsy & Behavior* **144**, pp. 109218. (2023). DOI: [10.1016/j.yebeh.2023.109218](https://doi.org/10.1016/j.yebeh.2023.109218)
7. Marie Ouellet and Sadaf Hashimi and **Vega Yon, George G.** "Officer Networks and Firearm Behaviors: Assessing the Social Transmission of Weapon-Use." *Journal of Quantitative Criminology*. (2022). DOI: [10.1007/s10940-022-09546-9](https://doi.org/10.1007/s10940-022-09546-9)
8. **George G. Vega Yon**. "Building, Importing, and Exporting GEXF Graph Files with rgexf." *Journal of Open Source Software* **6**, pp. 3456. (2021). DOI: [10.21105/joss.03456](https://doi.org/10.21105/joss.03456)
9. **Vega Yon, George G.** and Thomas, Duncan C. and Morrison, John and Mi, Huaiyu and Thomas, Paul D. and Marjoram, Paul. "Bayesian parameter estimation for automatic annotation of gene functions using observational data and phylogenetic trees." *PLOS Computational Biology* **17**(2), pp. 1-35. (2021). DOI: [10.1371/journal.pcbi.1007948](https://doi.org/10.1371/journal.pcbi.1007948)



10. Mohammadi, Mohammad and Najafzadehkhoei, Sima and **Vega Yon, George G.** and Wang, Yunshan. "A Novel Approach for Classifying Monoamine Neurotransmitters by Applying Machine Learning on UV Plasmonic-Engineered Auto Fluorescence Time Decay Series (AFTDS)." (). URL: <https://xlink.rsc.org/?DOI=D5NA00416K> DOI: 10.1039/D5NA00416K

Work in Progress and Technical Reports

1. **Vega Yon, George G.** and Thornton, Delaney and Redd, Andrew and Pulsipher, Andrew and Toth, Damon and Dorsan, Egenia and Rennert, Lior and Johnson, Kaitlyn and White, Laura F and Gruninger, Randon and Nolen, Leisha D and Samore, Matthew H. "Practical Guidelines and Reflections on Building Public Health Software: A Measles Case Study (Preprint)." (2026). DOI: [10.2196/preprints.94240](https://doi.org/10.2196/preprints.94240)
2. Love, Jay and LaPrete, Cormac R. and Sheets, Theresa R. and Vega Yon, George G. and Thomas, Alun and Samore, Matthew H. and Keegan, Lindsay T. and Adler, Frederick R. and Slayton, Rachel B. and Spicknall, Ian H. and Toth, Damon J.A.. "Characterizing Spatiotemporal Variation in Transmission Heterogeneity during the 2022 Mpox Outbreak in the USA." (2023). DOI: [10.1101/2023.05.10.23289580](https://doi.org/10.1101/2023.05.10.23289580)
3. **Vega Yon, George G.** and Pugh, Mary Jo and Valente, Thomas W.. "Discrete Exponential-Family Models for Multivariate Binary Outcomes." (2022).
4. **Vega Yon, George G.**. "Capital Necesario Unitario (CNU): Cálculo e Introducción Del Módulo de Stata CNU." (2014). URL: <https://ideas.repec.org/p/sdp/sdpwps/57.html>
5. Quintanilla, Ximena and Poblete, Isabel and **Vega Yon, George G.** and others. "Estudio Actuarial de los Fondos del Seguro de Cesantía." (2013).
6. Repetto, Andrea and **Vega Yon, George G.**. "El Impacto de un Alza en la Cotización Previsional: Pensiones, Salarios y Empleo." (2013).

Books

"Applied Network Science with R" (on development) <https://book.ggv.cl/>

"Applied HPC with R" (on development) <https://book-hpc.ggv.cl/>

Software Packages



1. Damon Toth and Jake Wagoner and Willy Ray and **George G. Vega Yon**. *multigroup.vaccine: Analyze Outbreak Models of Multi-Group Populations with Vaccination* (2026). R package version 0.1.1. URL: <https://epiforesite.github.io/multigroup-vaccine/>

downloads 251

2. Sima Najafzadehkhoei and **George G. Vega Yon** and Bernardo Modenesi. *epiworldRcalibrate: Fast and Effortless Calibration of Agent-Based Models using Machine Learning* (2026). R package version 0.1.0. URL: <https://github.com/sima-njf/epiworldRcalibrate>

downloads 142

1. **George G. Vega Yon**. *measles: Measles Epidemiological Models* (2025). R package version 0.1.1. URL: <https://github.com/UofUEpiBio/measles>

downloads 341

2. Damon Bayer and Dylan H Morris and **George G. Vega Yon** and Trevor Martin and Subekshya Bidari. *PyRenew: A Package for Bayesian Renewal Modeling with JAX and NumPyro* (2024). Python package version 0.1.0. URL: <https://github.com/CDCgov/PyRenew>

3. Kaitlyn Johnson and Dylan Morris and Sam Abbott and Christian Bernal Zelaya and **George G. Vega Yon** and Damon Bayer and Andrew Magee and Scott Olesen. *wwinference: Jointly infers infection dynamics from wastewater data and epidemiological indicators* (2024). R package version 0.1.1.99, <https://cdc.gov.github.io/ww-inference-model/>. URL: <https://github.com/cdcgov/ww-inference-model/>

4. **George G. Vega Yon**. *defm: Estimation and simulation of Multi-binary response models* (2023). R package version 0.1.0. URL: <https://cran.r-project.org/package=defm>

downloads 1694

5. Derek Meyer and **George G. Vega Yon**. *epiworldR: Fast Agent-Based Epi Models* (2023). R package version 0.0-2. URL: <https://cran.r-project.org/package=epiworldR>

downloads 12K

6. Derek Meyer and **George G. Vega Yon**. *epiworldRShiny: Shiny interface for epiworldR* (2023). R package version 0.1-0. URL: <https://cran.r-project.org/package=epiworldRShiny>

downloads 11K

7. **George G. Vega Yon**. *aphylo: Statistical Inference of Annotated Phylogenetic Trees* (2022). R package version 0.2-1. URL: <https://cran.r-project.org/package=aphylo>

downloads 16K

8. **Vega Yon, George G.** *epiworld: A Flexible and General Agent Based Model Engine* (2022). C++ library version 0.0-1. URL: <https://github.com/UofUEpiBio/epiworld>



9. **George G. Vega Yon**. *netplot: Beautiful graph drawing* (2021). R package version 0.1-1. URL: <https://cran.r-project.org/package=netplot>

downloads 20K
10. **Vega Yon, George G.** *rgexf: Build, Import and Export GEXF Graph Files* (2020). R package version 0.16.0. URL: <https://CRAN.R-project.org/package=rgexf>

downloads 649K
11. **George G. Vega Yon** and Thomas Valente. *netdiffuseR: Analysis of Diffusion and Contagion Processes on Networks* (2020). R package version 1.22.0. URL: <https://github.com/USCCANA/netdiffuseR>

downloads 54K
12. **Vega Yon, George G.** and de la Haye, Kayla. *ergmito: Exponential Random Graph Models for Small Networks* (2020). R package version 0.3-0. URL: <https://cran.r-project.org/package=ergmito>

downloads 27K
13. **Vega Yon, George G.** *slurmR: A Lightweight Wrapper for 'Slurm'* (2020). R package version 0.4-1. URL: <https://CRAN.R-project.org/package=slurmR>

downloads 36K
14. **Vega Yon, George G.** *fmcmm: A friendly MCMC framework* (2020). R package version 0.3-0. URL: <https://CRAN.R-project.org/package=fmcmm>

downloads 33K
15. **Vega Yon, George G.** *barry: your to-go motif accountant* (2020). C++ library version 0.0-1. URL: <https://github.com/USCbiostats/barry>

downloads 1030
16. **Vega Yon, George G.** *pruner: Implementing the Felsenstein's Tree Pruning algorithm* (2020). C++ library version 0.0-1. URL: <https://github.com/USCbiostats/pruner>
17. **George G. Vega Yon** and Brian Quistorff. *parallel: Stata Module for Parallel Computing* (2019). Stata Module version 1.20.0. URL: <https://github.com/gvegayon/parallel>
18. **Vega Yon, George G.** *googlePublicData: Working with Google's 'Public Data Explorer' DSPL Metadata Files* (2017). R package version 0.16.1. URL: <https://CRAN.R-project.org/package=googlePublicData>

downloads 59K
19. **George G. Vega Yon** and Muñoz, Enyelbert. *ABCOptim: Implementation of Artificial Bee Colony (ABC) Optimization* (2017). R package version 0.15.0. URL: <https://CRAN.R-project.org/package=ABCOptim>



downloads 90K

20. **George G. Vega Yon**. *twitterreport: Out-of-the-box analysis and reporting tools for twitter* (2016). R package version 0.16. URL: <https://doi.org/10.5281/zenodo.44528>

Conference talks/workshops

1. **George G. Vega Yon** and Thomas W. Valente and Jacob Kean and Mary Jo Pugh. "Expanding the ERGM Framework: Modeling Interrelated Health Outcomes with Jointly-Distributed Binary Data." In: Sunbelt Conference, 2025. (2025). (conference talk, [slides](#))
2. **George G. Vega Yon** and Anival Olivera and Thomas W. Valente. "Understanding Diffusion with netdiffuseR." In: Sunbelt Conference, 2025. (2025). (conference workshop, [slides](#))
3. Sima Najafzadehkhoei and **George G. Vega Yon**. "Automatic Calibration of Agent-Based Models using Deep Neural Networks." In: DELPHI Symposium 2024. (2024). (conference poster, [slides](#))
4. Aníbal Olivera and **George G. Vega Yon** and Chong Zhang and Alun Thomas and Matthew Samore and Karim Khader. "How does network structure in Agent-Based Models affect epidemiological parameters? A large." In: CSS 2024 Santa Fe. (2024). (conference poster, [slides](#))
5. **George G. Vega Yon** and Derek Meyer and Milo Banks and Matthew Samore. "epiworld: A fast multi-language library for epi ABMs." In: Conference on Complex Systems, 2024. (2024). (conference talk, [slides](#))
6. **George G. Vega Yon** and Matthew Samore and Karim Khader and Chong Zhang and Alun Thomas. "How Network Structure Affects Epidemiological Indicators in ABMs - A Large Simulation Study Featuring Artificial and Real-world Networks." In: INSNA Sunbelt 2023. (2023). (conference talk, [slides](#))
7. **George G. Vega Yon** and Derek Meyer. "Simulating Complex Agent-Based Model with epiworldR: A fast and flexible ABM framework." In: INSNA Sunbelt 2023. (2023). (conference workshop, [slides](#))
8. Porter Bischoff and **George G. Vega Yon**. "Visualization tools for network science: Building beautiful graphs with netplot." In: INSNA Sunbelt 2023. (2023). (conference poster, [slides](#))
9. Matthew Samore and **George G. Vega Yon** and Karim Khader and Kyle Kazemini and Nelson Chang and Lindsay Visnovsky and Alun Thomas and Candace Haroldsen and Kristina Stratford and Chong Zhang. "The importance of interaction networks in long-term care facilities to reduce the equilibrium prevalence of infectious pathogens." In: Sunbelt 2022. (2022). (conference talk)
10. Kyosuke Tanaka and **George G. Vega Yon**. "Imaginary Network Motifs: Structural Patterns of False Positives and Negatives in Social Networks." In: 72th Annual Conference of the International Communication Association. (2022). (conference talk)



11. **George G. Vega Yon**. "Una Mirada General a los Modelos Cuantitativos de Redes Sociales." In: VII Reunión Latinoamericana de Análisis de Redes Sociales. (2021). (conference talk, [slides](#))
12. Marie Ouellet and Sadaf Hashimi and **George G. Vega Yon**. "Network Influence in Officer Firearm Use." In: NETWORKS 2021 - A Joint Sunbelt and NetSci Conference. (2021). (conference talk)
13. **Vega Yon, George**. "Triads, Dyads, and Gene Functions - When Social Network Analysis Meets Phylogenetics." In: NETWORKS 2021 - A Joint Sunbelt and NetSci Conference. (2021). (conference talk)
14. **George G. Vega Yon** and Aileen Dinkjian and Sarah Hamm-Alvarez and Kayla de la Haye. "ERGMito Statistical Models for Small Team Social Networks." In: SciTS 2020. (2020). (conference talk, [slides/video](#))
15. **George G. Vega Yon** and Andrew Slaughter and Kayla de la Haye. "Exact Statistics and Semi-Parametric Tests for Small Network Data." In: IC2S2, 2019. (2019). (conference talk, [slides](#))
16. **George G. Vega Yon** and Andrew Slaughter and Kayla de la Haye. "Exact Statistics and Semi-Parametric Tests for Small Network Data." In: Sunbelt 2019. (2019). (conference talk, [slides](#))
17. **George G. Vega Yon** and Kayla de la Haye. "Small network statistics for the network science of teams." In: NetSciX 2019, SCL. (2019). (conference talk, [slides](#))
18. **Vega Yon, George**. "Computacion de Alto Rendimiento con R." In: satRday Santiago 2018. (2018). (conference workshop, [slides](#))
19. **George G. Vega Yon** and Kayla de la Haye. "Big Problems for Small Networks: Small Network Statistics." In: NASN 2018, DC. (2018). (conference talk, [slides](#))
20. **George G. Vega Yon** and Brian Quistorff. "Uncomplicated Parallel Computing with Stata." In: Stata Conference 2017. (2017). (conference talk, [slides](#))
21. **George G. Vega Yon** and Thomas W. Valente. "Understanding Diffusion with netdiffuseR." In: NASN 2017. (2017). (conference workshop, [slides](#))
22. **George G. Vega Yon** and Kayla de la Haye and Hee-sung Shin and Thomas Valente. "Diffusion of Smoking Initiation Among Diverse, Urban American Adolescents Over The High School Period." In: NASN 2017. (2017). (conference talk, [slides](#))
23. **George G. Vega Yon** and Thomas W. Valente. "Network Diffusion of Innovations in R: Introducing netdiffuseR." In: IC2S2, 2016. (2016). (conference poster, [slides](#))
24. **George G. Vega Yon** and Thomas W. Valente. "Network Diffusion of Innovations in R: Introducing netdiffuseR." In: user! 2016. (2016). (conference talk, [slides/video](#))
25. **George G. Vega Yon** and Thomas W. Valente. "Understanding Diffusion with netdiffuseR." In: Sunbelt Conference, 2016. (2016). (conference workshop, [slides/video](#))
26. **Vega Yon, George**. "Just tired of endless loops! or parallel: Stata module for parallel computing." In: Stata Conference, 2013. (2013). (conference talk, [slides](#))

Invited Speaker

1. ""Machine Learning Sandwich for Agent-Based Models: Automatic calibration and post-processing adjustment for ABMs"" In: Methods talks. (2024). (invited talk, [slides](#))
2. ""A Brief Introduction to High-Performance Computing: With Applications in R"" In: Methods talks. (2024). (invited talk, [slides](#))
3. ""Exponential Random Graph Models: An Introduction"" In: Summer Institute in Computational Social Science. (2024). (invited talk, [slides](#))
4. ""La Muerte del Aprendizaje de Máquina' o 'cómo la teoría puede mejorar a las máquinas"" In: Escuela de Verano de Sistemas Complejos. (2024). (invited talk, [slides](#))
5. "What is a Scientist?" In: Career Day 2023. (2023). (invited talk, [slides](#))
6. "Power and multicollinearity in small networks: A discussion of "Tale of Two Datasets: Representativeness and Generalisability of Inference for Samples of Networks" by Krivitsky, Coletti & Hens." In: Joint Statistical Meetings, 2023. (2023). (invited talk, [slides](#))
7. "Network visualization (in R) with "netplot" and motif counting (in C++) with "barry"." In: SCI Seminar. (2023). (invited talk, [slides](#))
8. "Prediction of Gene Functions by Leveraging Biological Insights with Mechanistic Machine Learning." In: University of Utah Data Science Seminar. (2023). (invited talk, [slides/video](#))
9. "Triadas, lazos y funciones genéticas: cuando las redes sociales y la filogenética se encuentran." In: Seminario de Data Science. (2021). (invited talk, [slides/video](#))
10. "Triads, Dyads, and Gene Functions - When Social Network Analysis Meets Phylogenetics." In: Computational Cancer Genomics Working Group Evening Lecture. (2021). (invited talk, [slides/video](#))
11. ""Predicción de funciones genéticas utilizando evidencia experimental y árboles filogenéticos: Un modelo evolutivo" o "Ciencia de datos en la práctica"." In: PUC Mathematical and Computational Engineering Seminars. (2020). (invited talk, [slides](#))
12. "A Brief Introduction to Using R for High-Performance Computing." In: Orange County R Users Group. (2019). (invited talk, [slides/video](#))
13. "Big Problems for Small Networks: Statistical Analysis of Small Networks and Team Performance." In: UCI Social Network Research Group. (2019). (invited talk, [slides](#))
14. "Big Problems for Small Networks: Statistical Analysis of Small Networks and Team Performance." In: SONIC Speaker. (2019). (invited talk, [slides/video](#))
15. "Estadística de Redes y Econometría Espacial (con R)." In: IMFD Summer School. (2018). (invited talk, [slides/video](#))

Other Talks

1. "Introduction to Exponential-Family Random Graph Models." In: 2nd Chilean Summer School about Social Network Research. (2026). (workshop, [slides](#))
2. "Statistical Inference in Network Science: A short course." In: 3ra Escuela de Verano en Ciencias de la Complejidad Social. (2026). (workshop, [slides](#))
3. "Introduction to Exponential-Family Random Graph Models." In: 1st Chilean Summer School about Social Network Research. (2024). (workshop, [slides](#))
4. "Statistical Inference in Network Science: A short course." In: Escuela de Verano de Sistemas Complejos. (2024). (workshop, [slides](#))
5. "Mechanistic Machine Learning." In: MInD2.0 Visit Harvard. (2023). (talk, [slides](#))
6. "A Crash course on git." In: Happy Scientist Seminar. (2021). (workshop, [slides/video](#))
7. "HPC with Slurm, R, and the slurmR R package." In: Happy Scientist Seminar. (2021). (workshop, [slides/video](#))
8. "Essays on Bioinformatics and Social Network Analysis: Statistical and Computational Methods for Complex Systems." In: Doctoral Defense. (2020). (talk, [slides](#))
9. "Essays on Bioinformatics and Social Network Analysis Statistical and Computational Methods for Complex Systems." In: Biostats Seminars. (2020). (talk, [slides/video](#))
10. "slurmR workshop." In: Happy Scientist Semminar Series. (2020). (workshop, [slides](#))
11. "Happy Scientist Seminar: Research Pipelines." In: Happy Scientist Seminar. (2019). (talk, [slides](#))
12. "What drives social networks? A gentle introduction to exponential random graph models (with a focus on small networks)." In: East LA R User Group. (2019). (talk, [slides](#))
13. "Overview of Social Network Models." (2018). (talk, [slides](#))
14. "A brief introduction to using R for high-performance computing." In: East LA R User Group. (2018). (talk, [slides](#))
15. "R Bootcamp for Scientific Computing 2018." In: R Bootcamp for Scientific Computing. (2018). (workshop, [slides](#))
16. "Introduction to R (for HPC users)." In: USC's HPCC workshop. (2018). (workshop, [slides](#))
17. "Intro to R." In: USC's HPCC workshop. (2018). (workshop, [slides](#))
18. "Reproducible Research." In: Health Behavior Research Students talk. (2016). (talk, [slides](#))

Teaching

(PHS 7045) Advanced Programming with R and HPC

Fall 2024

The University of Utah, USA

Co-instructor, PHS Ph.D. Program



(PHS 7045) Advanced Programming with R and HPC

Fall 2022

The University of Utah, USA

Co-instructor, PHS Ph.D. Program

(PM 566) Introduction to Health Data Science

Fall 2021

University of Southern California, USA

Instructor, Masters of Science in Public Health Data Science

(PM 566) Introduction to Health Data Science

Fall 2020

University of Southern California, USA

Co-instructor, Masters of Science in Public Health Data Science

Statistical Computing with Stata

First semester 2012

Universidad Adolfo Ibáñez, Chile

Instructor, Masters in Economics and Public Policy

Introduction to Economics

First semester 2012

Universidad Adolfo Ibáñez, Chile

Co-instructor, B.A. in Business Administration

Microeconomics

Second semester 2012

Universidad Adolfo Ibáñez, Chile

Co-instructor, B.A. in Business Administration

Introduction to Economics

First semester 2011

Universidad Adolfo Ibáñez, Chile

Co-instructor, B.A. in Business Administration

Mentoring/Advising

Mentor, RA (2024–Present)

Sima Najafzadehkhoei, University of Utah, Population Health Sciences, Ph.D. in Biostatistics.

Computational Epidemiology, Agent-Based Models and ML.

Mentor, RA (2024–Present)

Anibal Olivera, Universidad del Desarrollo, Chile, Ph.D. in Social Complexity. Network Science and R package development.



Mentor, RA (2024–Present)

Milo Banks, Rowland Hall High School, High School Student. Software development and computational epidemiology.

Mentor, RA (2023–Present)

Hyrum Thomas Diesen, University of Utah, School of Biological Sciences, Molecular Cellular Evolutionary Biology Ph.D. program.

Ph.D. Dissertation Committee Member (2023–Present),

Eric Anto, University of Utah, Population Health Sciences, Biostatistics Ph.D. program.

Qualifying Exam chair and Member, Ph.D. Dissertation Committee Member (2022–2024),

Katherine Lawson Michod, University of Utah, Population Health Sciences, Clinical and Translational Epidemiology Ph.D. program.

Mentor, RA (Summer 2023),

Porter Bischoff, University of Utah + Utah Valley University, Summer Program for Undergraduate Research (SPUR), Network Visualization using R.

Supervisor, RA, and M.Sc. Dissertation Committee Member (2022–Present)

Derek Meyer, University of Utah, Population Health Science, Agent-Based Models for Epidemics.

Mentor, (2022–Present),

Jacqueline M. Kent-Marvik, University of Utah, Nursing School, Ph.D. Health Sciences, Network Science and Social Network Analysis.

Mentor, (2022),

Luis Lopez, NIH, Postbac program, Exponential-Family Random Graph Models.

Grants

(PIs: Vega Yon, Basu Roy)

University of Utah Seed Grant

2026 – 2027

Quantum-Enhanced Epidemiological Modeling for Public Health Decision-Making

The project seeks to explore the potential of quantum computing to enhance epidemiological modeling and public health decision-making. By implementing a quantum-enhanced version of an epidemiological ABM, we aim to demonstrate the feasibility and advantages of quantum computing in this domain, as well as to provide insights into the types of epidemiological problems that could



benefit most from quantum approaches.

Role: Co-PI

Total Award Amount: \$ 50,000.

5P01CA196569-08 (PIs: Gauderman, Siegmund)

07-01-2016 – 08-31-2027

National Cancer Institute (NCI)

0.6 calendar months/year

Statistical Methods for Integrative Genomics in Cancer

This program aims to develop novel statistical methods for integrating multi-omic data to address cancer etiology, prognosis, and treatment. The program has three synergetic projects, one using evolutionary models for annotating genes and their products (of which I am a Co-I).

Role: Co-Investigator

Total Award Amount (including indirect costs): \$ 12,894,663.

HT94252310221 (PI: Kennedy)

07-01-2023 – 30-06-2026

U.S. Army Medical Research and Development Command (DoD)

1.8 calendar months/year

Phenotypes of Epilepsy Etiology and Drug Resistance (PEER)

The project aims to investigate the causes and workings of epilepsy by utilizing new data science methods. The grant will be utilized to analyze patterns in three ongoing and established studies, generating “risk scores” for epilepsy after a head injury, specifically among military personnel.

Role: Co-Investigator

Total Award Amount (including indirect costs): \$ 846,484.

5U01CK000675-01 (PI: Keegan)

09-30-2022 – 09-29-2025

National Center for Emerging and Zoonotic Infectious Diseases (NCEZID) 0.3 calendar months/year

TRANSMIT: Training Research Acumen in Students Modeling Infectious Threats

A fellowship program oriented to train students in using mathematical modeling and computational methods to study infectious diseases.

Role: Co-Investigator

Total Award Amount (including indirect costs): \$ 891,356.

Submitted but not funded

1R01HG012878-01 (PI: Vega Yon)

12-01-2022 – 11-30-2027

National Human Genome Research Institute (NHGRI)

Improving our Predictive Capability of Gene Functions by Leveraging Biological Insights with Advanced Statistical Computing

Using discrete exponential-family models, build a mechanistic gene function evolution model incorporating complex features involving biological processes such as neofunctionalization.

Role: PI

Total Award Amount: Not funded



1R01HG012878-01A1 (PI: Vega Yon)

12-01-2023 – 11-30-2028

National Human Genome Research Institute (NHGRI)

Building a Novel Prediction Framework Leveraging Biological Insights to Boost Machine Learning Algorithms for Annotating Gene Function

This project extends my model of function evolution using discrete exponential-family models. It embeds it into a mechanistic machine learning framework—mixing theory and data-driven models to improve gene function prediction.

Role: PI

Total Award Amount: Not funded

Honors and Services to the Profession

Manuscript Review (Ad Hoc)

Journal of the American Statistical Association

American Sociological Review

The Official Journal of The Society for Computational Economics

The R Journal

The Stata Journal

Social Networks

Journal of Mathematical Sociology

Computer Methods and Programs in Biomedicine Update

Journal of Open Source Software

Bioinformatics

Abstract Review

International Conference on Computational Social Science (2019–2021)

SUNBELT Conference (2016)

Panelist/Discussant

“Learning tips for your PhD from other researchers in the community”.

Early and Middle Career Researchers (EMCRs) on Social Networks workshops, October 2023.

“JASA A&CS Special Invited Session: A Tale of Two Datasets – Representativeness and Generalisability of Inference for Samples of Networks.”

Joint Statistical Meetings (JSM), August 2023.

“Social Network Diffusion of Individual Behavior Change Interventions Virtual Workshop.”

National Institute on Aging Division of Behavioral and Social Research (NIA BSR), March 2022.

Book Review



"Microeconometrics and Matlab: An Introduction", by Adams, Clarke and Quinn, Oxford University Press, 2015.

"Mastering Gephi Network Visualization", by Ken Cherven, Packt Publishing, 2015.

"Network Graph Analysis and Visualization with Gephi", by Ken Cherven, Packt Publishing, 2013.

Misc

Core member of the "University of Utah's Center for Data Science" (2023–Present)

Lead of "Network Science and Social Network Analysis at the U (NetSNAU)" research group (2022–Present)

Member of "Center for Applied Network Analysis (CANA) at USC" research group (2016–Present)

Co-organizer of the USC Networks Meeting (2020, 2021)

Founder of the (first) R Users Group in Chile (2013)

Co-organizer of the East LA R User Group (LAERUG).

Software

R, C++, Python, Typst, SQL, XML, regex, Stata+Mata, VBA, Gephi, Pajek, Mathematica, MS Suite, Git, Unix, Docker, Visual Studio Code

last update: April 04, 2026

<https://ggvy.cl>